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The study combines two techniques: anther culture and molecular marker in selected salinity tolerance varieties on rice (Oryza Sativa L).

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Although salt stress is among the most important constraints to rice yield, few rice breeding programs try to develop cultivars with higher yield under salt stress. This research has shown considerable doubled haploid (DH) population was established via anther culture of C53/ Pokkali, C53/ Doc Phung and C53/D51 on N6 + 2mg/l 2,4D + 60g/lít Sucroz medium, which had been shown particularly suitable for anther culture of indica hybrids.Among these factors, the most important one is the genotypic difference. The results found that different rice species, or crosses quite differently in response to anther culture . On the other hand, culture medium also strongly influences the anther culture response. For example, N6 medium proved to be much more suitable for rice than for indica rice in anther culture.N6 complement very good for (20%) on C53/Pokkali, 10,5% for C53 / Doc Phung and only 4,50% on C53/D51 . Hovever the regenetaion for development gametoclonal on MS complement with 2mg/L BA+2mg/L kinetin. A total of 72 lines from anther culture were genotyped by a PCR method to identify presence/absence of salt genes . Some lines salt tolerance such as C53/■■c ph■ng -17, C53/■■c ph■ng -19, C53/pokkali -5, C53/pokkali -11, C53/pokkali -27, C53/pokkali -42, C53/pokkali -43, C53/pokkali -44, C53/D51 such as : C53/D51 -4, C53/D51 -5 và C53/D51 -8. This method shows that the ability to predict the resistance genotype and resistance phenotype is very high, so it can be applied to select resistant varieties for salty conditions, as a source of hybrid material for current new rice breeding programs.